

Introduction to Planning

Video Lesson

Duration: 00:14:05

Slide Deck

M10S01 (<https://canvas.tamu.edu/courses/430127/files/92570683?wrap=1>) 

Lesson Transcript

In this lecture and the three following lectures, we will look into planning.

Planning

AI lecture (Yoonsuck Choe): Materials from Russel and Norvig.

- 7.2, 7.3, 7.7.1 (3rd ed, 4th ed); : Wumpus world (an example domain), frame problem
- 10.4.2 (3rd ed); chapter 11 (4th ed): Situation calculus

The materials will be from our textbook, Russell and Norvig.

The corresponding chapters and sections for the third edition and for the fourth edition are shown below.

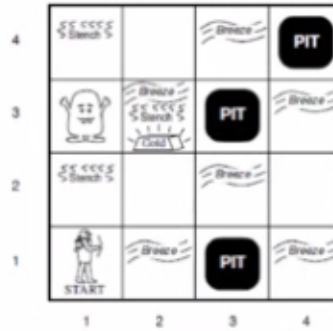
Planning

- The task of coming up with a sequence of actions that will achieve a goal is called **planning**.
- Simple approaches:
 - Search-based
 - Logic-based
- **Representation** of states and actions become important issues.

What is planning? In AI, the task of coming up with a sequence of actions that will achieve a goal is called planning. There are simple approaches based on search and also based on logic.

In any planning approach, the representation of states and actions become important issues.

Example Domain: Wumpus World



- Want to get to the gold and grab it.
- Want to avoid pits and the "wumpus".
- Clues: breeze near pits and stench near the wumpus.
- Other sensors: wall (bump), gold (glitter), kill (scream)
- Actions: move, grab, or shoot.

Let us consider planning in the context of this example domain called the Wumpus World.

Here is the agent at the grid location [1, 1]. You want to move this agent to the grid where there is gold and grab it.

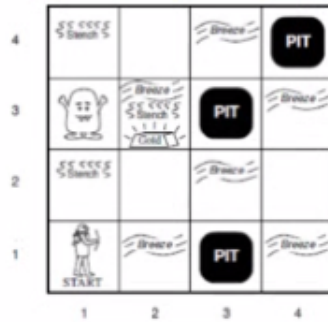
While doing so, you want to avoid the monster, the "wumpus," and also this pit.

How can you avoid the wumpus or the pit? When you move into a certain grid location, you get some kind of hint. For example, when there is a wumpus in the neighboring grid, there will be a stench. Also, when you are near a pit, you can feel the breeze.

There are other sensors, such as the wall. If you bump into the wall, you cannot proceed any further. Or when you are in the grid with the gold, then you perceive glitter. And as I will mention later on, you can also shoot an arrow. And when you hit the wumpus, you can hear the scream.

There are three basic actions: move between the grids, grab something, or shoot.

Wumpus World (WW)



Performance measure

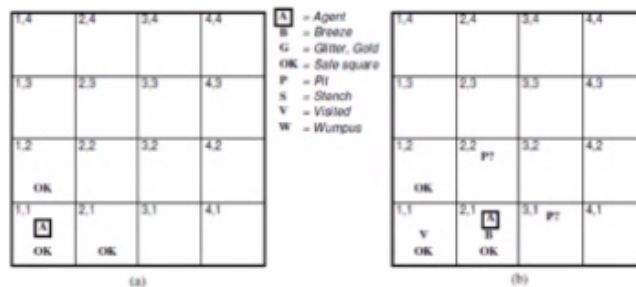
- +1000: picking up gold
- -1000: fall in a pit, or get eaten by the wumpus
- -1: each action taken
- -10: each arrow used



The goal is to maximize the performance, and the performance measure is positive 1000 when you pick up the gold and negative 1000 if you fall in the pit or get eaten by the wumpus. And there is a small penalty for taking each action.

As for the shooting of the arrow, there is a minus 10-point cost.

Evolution of Knowledge in WW



- Move from [1,1] to [2,1].
- Based on the sensory data (breeze: marked **B**),
 - we can mark [2,2] and [3,1] as potential pits (marked **P?**),
 - but no need to mark [1,1] as such since we came from there and we already know there's no pit there.

Let us consider a very simple exploration scenario. The agent is initially at grid location [1, 1]. There is no stench or breeze that is perceived. So, the agent can conclude that these two neighboring grids are safe. So, we mark them as okay, and since you are already in grid location [1, 1] and nothing is happening, you are also okay.

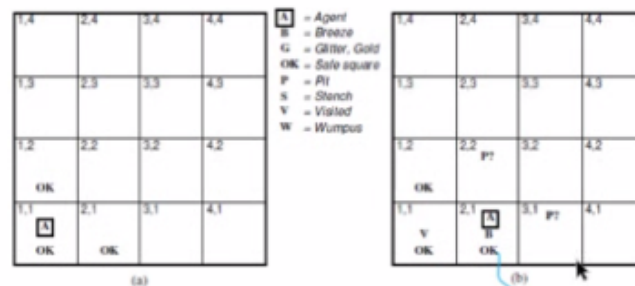
Based on this information, let us say the agent decided to move from grid location [1, 1] to [2, 1]. So, that is moved from [1, 1] to [2, 1]. And the agent's location now is at the grid location [2, 1].

Once the agent is at this grid location, it can sense what is around it. Here, because there was a pit in [3, 1], it feels the breeze.

But that is the only information that the agent has so far. Meaning that there is a breeze. So, there could be a pit either in [1, 1], [2, 2], or [3, 1]. But based on your prior knowledge that you are already in [1, 1], and it was okay, you can mark it off as okay. Whereas you do not know whether [2, 2] or [2, 3], or both contain a pit.

So, you mark it as **P?** to indicate that there is a potential pit in these two locations.

Evolution of Knowledge in WW



- Move from [1,1] to [2,1].
- Based on the sensory data (breeze: marked B),
 - we can mark [2,2] and [3,1] as potential pits (marked P?).
 - but no need to mark [1,1] as such since we came from there and we already know there's no pit there.

One thing to note here is that in grid location [2, 1], there is no stench, and this information will turn out to be useful in the next step. So, you should remember that for now.

Evolution of Knowledge in WW



- Move back to [1,1] and then to [1,2]: we can infer that the wumpus is in [1,3]
 - In [1,2], stench **S** is perceived: Wumpus can be in [1,1], [2,2], [1,3].
 - Grid [1,1] is okay since we've been there.
 - Grid [2,2] is okay, since we were at [2,1] and there was no stench **S**.
 - So, the Wumpus must be in [1,3].
 - Other facts: no breeze **B** at [1,2], so [2,2] is **OK** (it was previously **P?**).
- Then move to [2,2] and then to [2,3] where the gold can be found (glitter).

Now, let us consider this situation. So again, you went from [1, 1] to [2, 1], and then went back to [1, 1], and then it went up to [1, 2]. So, this is the sequence of actions so far.

Now, once the agent is at the grid location [1, 2] or the grid location [1, 2], it perceives the stench, **S**.

Just based on this knowledge, the agent can infer there could be a wumpus in [1, 3], [2, 2], or [1, 1]. So, these are the neighboring grids.

Evolution of Knowledge in WW



- Move back to [1,1] and then to [1,2]: we can infer that the wumpus is in [1,3]
 - In [1,2], stench **S** is perceived: Wumpus can be in [1,1], [2,2], [1,3].
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 - So, the Wumpus must be in [1,3].
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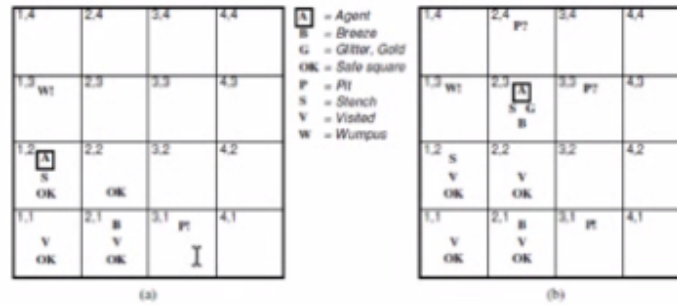
As shown here, these three could potentially have a wumpus in them. However, grid location [1, 1] is okay because we just came from that location, and we know that it is okay. So, we check off that location and can assert that it does not contain the wumpus.

Now the second case is a little bit complicated. Grid location [2, 2] is okay. Actually, the reason is while you were in the grid location [2, 1], there was no stench. Recall what I just asked you to remember in the previous slide. Because there is no stench here, there could not be any wumpuses in any of these three locations.

So with that, we can check this off.

With that, we can now make sure that grid location [1, 3] actually has the wumpus, so we mark it as **W!** to indicate that this is a correct fact.

Evolution of Knowledge in WW



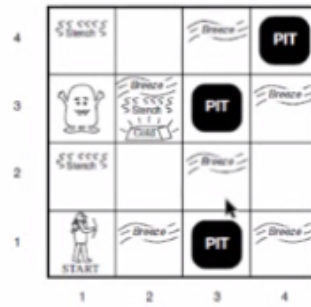
- Move back to [1,1] and then to [1,2]: we can infer that the wumpus is in [1,3]
 - In [1,2], stench **S** is perceived: Wumpus can be in [1,1], [2,2], [1,3].
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 - So, the Wumpus must be in [1,3].
 - Other facts: no breeze **B** at [1,2], so [2,2] is **OK** (it was previously **P?**).
- Then move to [2,2] and then to [2,3] where the gold can be found (glitter).

We can also derive other facts based on this perception. Because when you are at the grid location [1, 2], there is no breeze, so there cannot be a pit in the grid location [2, 2]. So, we can now mark it as "okay." And with that, this uncertainty of whether grid location [3, 1] has a pit or not is resolved. And now we can say that this grid location definitely has a pit.

So, that is **P!**

After asserting that these four locations are all safe, let us say you went from [1, 2] to [2, 2], and there is no perception of breeze or stench. You can move either to this direction or that direction, and let us say you decide to move to [2, 3]. Then you will feel the glitter and hold onto the gold.

Inference in Wumpus World



- Knowledge Base: basic rules of the Wumpus World.
- Additional knowledge is added to the KB: facts you gather as you explore ([x,y] has stench, breeze, etc.)
- We can ask if a certain statement is a logical consequence of the KB: "There is a pit in [1,2]"

In the previous slide, what we have seen is that we can do some kind of reasoning based on the different facts that we gather throughout our exploration.

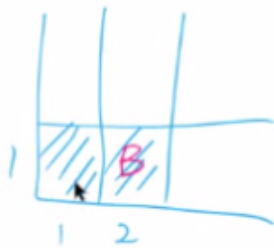
This is possible because we know the basic rules of the Wumpus World. For example, when there is a stench, then there is a wumpus nearby. When there is a breeze, there is a pit nearby.

Now, you can think of this in terms of a knowledge base. For example, for the Wumpus World, we will initially populate the knowledge base with the basic rules of the Wumpus World, but the agent does not have any knowledge about where the Wumpus is, where the pit is, or where the gold is. It has to discover these facts by moving around and associating certain perceptions like stench or breeze with a specific grid location.

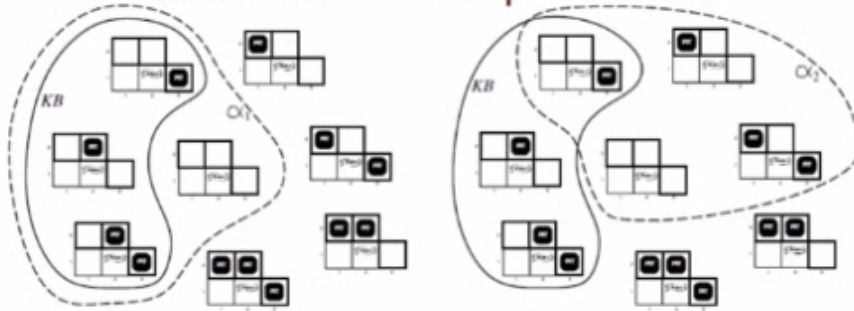
So, as the agent explores, additional knowledge is added to the knowledge base. For example, grid location [1, 2] contains stench. Grid location [2, 1] has this breeze, and so on.



, based on the initial knowledge base with the rules and the pieces of knowledge that we accumulate throughout exploration, then we can ask if a certain statement is a logical consequence of the knowledge base. For example, there is a pit in [1, 2] or not.



Inference in Wumpus World

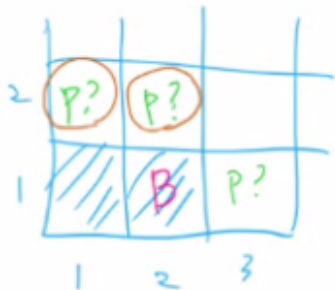


KB: basic rules, plus [1,1] and [2,1] explored

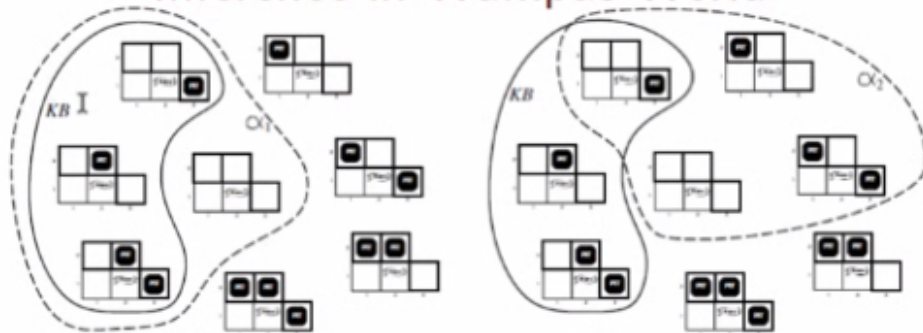
Pits in three unvisited grids [1,2], [2,2], [3,1]: $2^3 = 8$ possible cases

- α_1 = "There is no pit in [1,2]"
- α_2 = "There is no pit in [2,2]"
- Only α_1 follows from the KB (KB entails α_1 : model of KB is the subset of α_1), but not α_2 .
 - $KB \models \alpha_1$ (all models of KB satisfy α_1)
 - $KB \not\models \alpha_2$

Let us consider the case where the knowledge base includes all the rules and the facts associated with the grid location [1, 1] and [2, 1] being visited, and this includes that [1, 1] is okay, [2, 1] is okay, and then, grid [2, 1] has breeze in it.



Inference in Wumpus World



KB: basic rules, plus [1,1] and [2,1] explored

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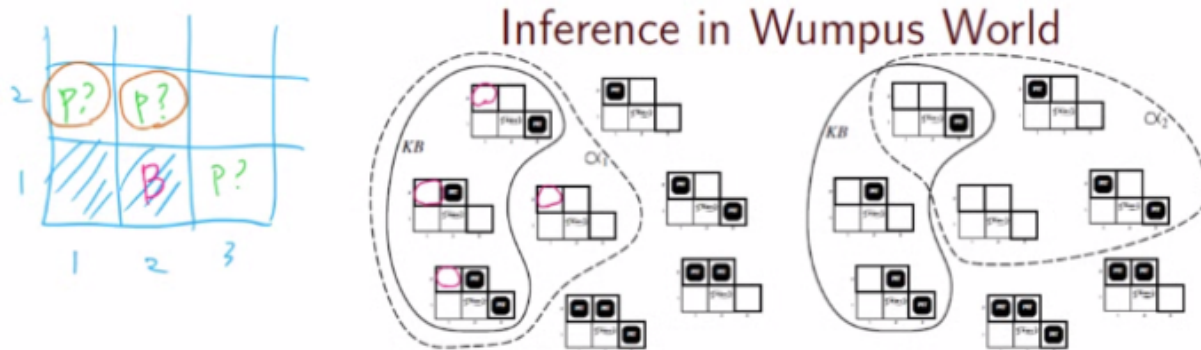
Given this, we want to find out whether there is a pit in [1, 2] or [2, 2].

If you enumerate all of the possibilities of there being a pit or not in these three locations, you

have 2^3 possible cases.

Of these eight possible scenarios, only these three comply with what we know in the knowledge base. Again, the knowledge base includes the rules and the fact that you visited these two locations, [1, 1] and [2, 1], and that there is a breeze in grid location [2, 1].

The question is whether you can infer this fact α_1 , "there is no pit in [1, 2]," based on your current knowledge.



KB: basic rules, plus [1,1] and [2,1] explored

Pits in three unvisited grids [1,2], [2,2], [3,1]: $2^3 = 8$ possible cases

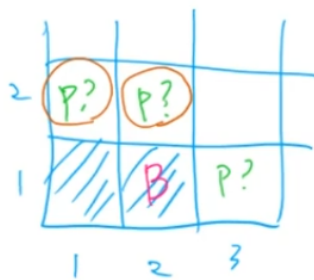
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One way of doing that is to find these cases that satisfy α_1 . That means there is no pit in [1, 2]. If you mark all of these where there is no pit in [1, 2], it will form this set.

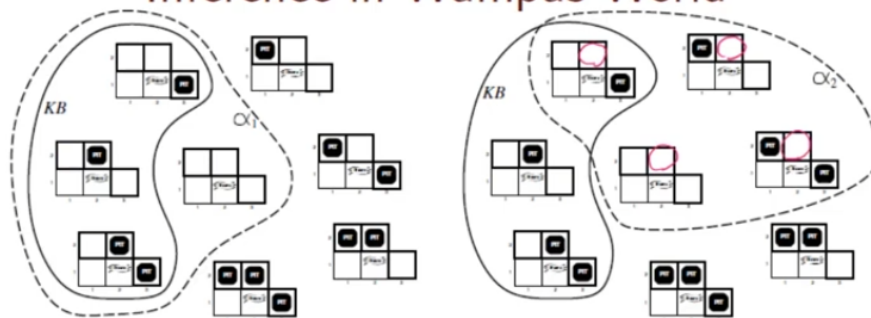
In this situation, we can find out whether α_1 logically follows from the knowledge base just by checking whether all of these cases that satisfy the knowledge base also satisfy α_1 .

In this case, you can easily verify that by checking whether this set contains this set or otherwise, whether the set that corresponds to the knowledge base is a subset of the set that corresponds to α_1 , and in this case, that happens to be the case.

So, we can say that we can assert that there is no pit in [1, 2] based on your knowledge in the knowledge base.



Inference in Wumpus World



KB: basic rules, plus [1,1] and [2,1] explored

Pits in three unvisited grids [1,2], [2,2], [3,1]: $2^3 = 8$ possible cases

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However, this is not the case for α_2 , which says there is no pit in [2, 2]. Again, we can look at all of these eight cases and find the ones where there is no pit in the grid location [2, 2]. So, that would be these four.

But as you can see, the set for α_2 does not contain the set for the knowledge base. So, we can say that we can assert α_2 is true based on our knowledge and the knowledge base.

Formally, we say in this case that the knowledge base entails α_1 . In this case, it means that all models of the knowledge base, the examples that satisfy the knowledge base, also satisfy α_1 .

As we have seen, the knowledge base does not entail α_2 .

Propositional-logic-based Agent

- Query KB: Is there a Wumpus in [x,y]? Is there a pit in [x,y]?
- Add knowledge to KB (perceptual input): Breeze felt in [x,y], Stench detected in [x,y], etc.
- Decide which action to take (move where, etc.): Move to [x,y], grab gold, etc.

Note: here, there's only one goal, to grab the gold. Can we specify an arbitrary goal and derive a plan?

Problem: Propositions need to be explicit about location, e.g., $Breeze_{x,y}$, $Stench_{x,y}$, $\neg Wumpus_{x,y}$.

Ultimately, we want to issue queries like this. Is there a wumpus in location [x, y]? Is there a pit in location [x, y]? And so on. And our first attempt would be just to use propositional logic.

For example, when we explore the environment, we can add new knowledge to the knowledge

base based on perceptual input. For example, there is a breeze at $[x, y]$, stench at $[x, y]$, et cetera. And based on that, we want to take action, move to $[x, y]$ and grab the gold, and so on.

The problem with this approach is that you must make the $[x, y]$ location explicit as part of the symbol or the atoms in the propositional logic formula.

For example, to represent that there is a breeze at location $[x, y]$, you must have a single atom containing x, y as part of its name. Same thing for the stench. You have to have stench x, y as a single atom.

And for the wumpus, again, you have to have wumpus x, y as a single atom.

Also, there are other considerations. For example, in our example, there is only one goal, which is to grab the gold. But can we specify an arbitrary goal and derive a plan? For example, shoot the wumpus to kill it.

Due to these difficulties, we will at least need something close to first-order logic.

And that is the topic of our next lecture.